Face-Rotation BFS Nets of the Regular and Uniform Convex 4-Polytopes

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Abstract

We study the face-rotation BFS unfolding of regular convex 4-polytopes: a breadth-first spanning tree of the cell-adjacency graph is chosen, and each child cell is rotated about the shared polygonal face into the parent's 3-dimensional hyperplane. We prove by exhaustive computational verification that for every one of the six regular convex 4-polytopes—the 5-cell, 8-cell, 16-cell, 24-cell, 120-cell, and 600-cell—this procedure yields a valid (non-self-intersecting) 3D net from *every* choice of root cell. In total, $5 + 8 + 16 + 24 + 120 + 600 = 773$ root cells were verified. For the 5-, 8-, 16- and 24-cell we verify more: *every* BFS spanning tree of every root is valid, 1,118,261 trees in all, so for those four the conclusion does not depend on how ties in the traversal are broken. This extends the work of Devadoss and Harvey [3], who established validity for the 5-, 8-, and 16-cell for *all* spanning trees and did not settle the remaining three; we resolve them for the BFS subfamily. That subfamily is exactly the right one. None of those three polytopes is *all-net*: each admits some spanning tree whose unfolding overlaps [6, 4], so the classification of the property is now complete and the three we add are exactly the three that fail it. Yet all 786,432 BFS trees of the 24-cell are valid, and none of the known counterexamples is a BFS tree. Our result stands in sharp contrast to the apple-peel unfolding algorithm [10], for which the 120-cell produces no valid nets despite admitting 1,440 valid orderings, and the 600-cell admits no valid ordering at all. We also show that regularity is essential. Running the same procedure from all 24,487 roots of the 64 convex uniform 4-polytopes, the outcome comes to depend on which spanning tree is taken, in a way it never does in the regular case: the grand antiprism, for one, is valid from all 320 of its roots under one rule for choosing the tree and from none of them under another. The runcinated 24-cell is sharper still: no index-based rule we tried yields a valid net from any of its 240 roots, and yet a tree chosen equivariantly yields one from every root.

1 Introduction

The question of whether a convex polytope can be *unfolded* into a non-overlapping flat arrangement of its facets—a *net*—is a classical problem in discrete geometry, dating back to Dürer's depiction of polyhedral nets in the sixteenth century. In three dimensions the analogous question (does every convex polyhedron admit a non-overlapping edge unfolding?) remains open as the Dürer conjecture [7]. For four-dimensional polytopes, the facets are 3-dimensional cells and a net is an arrangement of those cells in $\mathbb{R}^3$. Non-overlap is not a cosmetic condition: a point covered twice would have to travel to two different cells of the polytope when the net is folded up, so only an injective unfolding can be cut from a single solid body (Remark 1). Early work on unfolding the 4-cube into 3D arrangements of its cubical cells was carried out by Turney [8], who enumerated all 261 topologically distinct nets of the 8-cell (hypercube).

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Devadoss and Harvey [3] initiated the systematic study of ridge unfoldings of the six regular convex 4-polytopes, proving that every spanning tree of the cell-adjacency graph yields a valid net for the 4-simplex (5-cell), 4-cube (8-cell), and 4-orthoplex (16-cell). They demonstrated failure for the analogous result in dimension $n \geq 5$ for orthoplexes, and showed that the 600-cell too admits an overlapping unfolding; the 24-cell and the 120-cell have since been shown to admit one as well [6, 4].

In this paper we study a canonical subfamily of spanning trees, namely those produced by *breadth-first search* (BFS) from a chosen root cell, paired with a specific unfolding procedure that we call *face-rotation*. Our main result (Theorem 2) is that for every regular convex 4-polytope and every choice of root cell, this procedure yields a valid 3D net. The proof is by exhaustive computational verification. The result for the 5-, 8-, and 16-cell is a special case of [3]; the new content is the 24-cell, 120-cell, and 600-cell.

A companion paper [10] introduces the *apple-peel* algorithm, a greedy spiral/zonal ordering of cells that defines a different family of spanning trees. That algorithm completes successfully for all 1,440 starting pairs for the 120-cell, yet every resulting 3D net self-intersects. For the 600-cell the algorithm cannot even complete the ordering. The face-rotation BFS algorithm succeeds where apple-peel fails, illustrating that the existence of a valid ordering and the validity of a net are independent properties.

This paper is organized as follows. Section 2 recalls the necessary background and describes the face-rotation BFS algorithm. Section 3 states and verifies Theorem 2 and discusses the contrast with apple-peel unfolding. Section 4 runs the same procedure over the 64 convex uniform 4-polytopes and shows that the theorem does not survive the loss of regularity. Section 5 collects open problems.

2 Preliminaries

2.1 Background

There are exactly six regular convex 4-polytopes (see Coxeter [1]). Their Schläfli symbols and combinatorial data are listed in Table 1.

Table 1: The six regular convex 4-polytopes.

Polytope	Symbol	Vertices	Faces	Cells	Cell type	Cell degree
5-cell	$\{3,3,3\}$	5	10	5	tetrahedron	4
8-cell	$\{4,3,3\}$	16	24	8	cube	6
16-cell	$\{3,3,4\}$	8	32	16	tetrahedron	4
24-cell	$\{3,4,3\}$	24	96	24	octahedron	8
120-cell	$\{5,3,3\}$	600	720	120	dodecahedron	12
600-cell	$\{3,3,5\}$	120	1200	600	tetrahedron	4

Here *cell degree* denotes the number of cells adjacent to each cell (sharing a ridge, i.e., a 2-face).

Following Devadoss and Harvey [3], a *ridge unfolding* of a 4-polytope P is specified by a spanning tree T of the cell-adjacency graph $G(P)$ (nodes = cells, edges = shared ridges, i.e., shared 2-faces). The unfolding is performed inductively: a root cell is placed in some 3D hyperplane H; for each edge (p, c) in T directed away from the root, the child cell c is rigidly rotated about the shared polygon with its parent p until it lies in the same hyperplane as p. The resulting arrangement is a *net* if no two cells overlap in their interiors.

Remark 1 (Why overlap must be excluded)**.** *The map from the boundary ∂P of P (the union of its cells) into H induced by the construction above is well defined whether or not cells overlap, and running the rotations backwards reassembles P in either case. What overlap destroys is the* inverse *assignment. If two cells overlap, a point of the overlap belongs to both of them, and folding sends it to two distinct cells of P; a solid body occupying that region of H therefore has no consistent destination, since one piece of matter cannot be in two places of $\mathbb{R}^4$ at once. Equivalently, an overlapping arrangement is an immersed image of ∂P rather than a subset of H, so it cannot be cut from a single solid block in the first place. Requiring the unfolding to be injective is thus the exact analogue of requiring a polyhedral net to be cuttable from one sheet of paper, and it is why validity, rather than the mere existence of an unfolding, is the object of study here.*

Ridge unfoldings are also accessible through interactive software. Stella4D [9] has displayed 3D nets of polychora since 2007 and lets the user reglue cells one at a time, so that in principle any spanning tree can be realized by hand. Its nets are generated to maximize symmetry and visual appeal rather than to realize a prescribed family of spanning trees, and, as its documentation states, intersections between cells are ignored. Whether an unfolding is free of self-intersection—the question studied here—is therefore outside the scope of such tools.

Definition 1 (BFS spanning tree)**.** *Let $G(P)$ be the cell-adjacency graph of a 4-polytope P, let r be a root cell, and write L_k for the set of cells at distance k from r in $G(P)$. A* BFS spanning tree *rooted at r is a spanning tree in which the parent $p(v)$ of each cell $v \neq r$ is one of its* candidates*, the cells of $N(v) \cap L_{d(v)-1}$, where $N(v)$ is the set of neighbors of v in $G(P)$ and $d(v)$ is the distance from r to v. The layers L_k are determined by P and r alone; a BFS tree is exactly a choice of such a parent for each cell, and different choices give different trees. Equivalently, a spanning tree rooted at r is a BFS tree if and only if every path in it from r is a shortest path in $G(P)$. We use the name for this family of trees, not for any one traversal: the definition above mentions no order in which cells are visited. Note also that $G(P)$ itself does not depend on the root: the root selects a tree in it.*

Lemma 1 (What a traversal can reach)**.** *A BFS tree rooted at r is produced by some breadth-first traversal if and only if there is a total order $\prec$ on the cells with*

$$p(v) \;=\; \min\nolimits_{\prec}\big(N(v) \cap L_{d(v)-1}\big) \qquad \text{for every } v \neq r,$$

and such an order exists if and only if the digraph with an edge $p(v) \to q$ for every other candidate q of every v is acyclic.

Proof. A traversal handles the cells of a layer in the order it dequeues them, and v is claimed by the first of its candidates to be dequeued; that dequeue order is a total order, which gives one direction. Conversely, given $\prec$, process the layers in turn and each layer in increasing $\prec$: then v is claimed by its $\prec$-least candidate. The order must satisfy $p(v) \prec q$ for every candidate $q \neq p(v)$ of every v, and a total order meeting a set of such constraints exists exactly when they have no cycle. Since all candidates of v lie in $L_{d(v)-1}$, the digraph splits by layer. □

Remark 2 (The family is larger than what a traversal reaches)**.** *Lemma 1 makes the gap between the two readings of "BFS tree" measurable, and it is wide. Table 2 counts, for the polytopes of part 1 of Theorem 2, how many of the trees allowed by Definition 1 a traversal can reach. The counts are the same from every root, as cell-transitivity requires.*
The obstruction is not the obvious one. If two cells of a layer share two candidates, no traversal can give them different parents from that pair; but the 16-cell has no such pair from any root and still leaves three quarters of its trees unreachable, because the constraints can close a cycle without

Table 2: BFS trees per root, and how many of them a breadth-first traversal can produce.

Polytope	BFS trees	Reachable	
5-cell	1	1	100%
8-cell	6	6	100%
16-cell	20,736	4,896	23.6%
24-cell	32,768	2,592	7.9%

any two cells sharing two candidates—v_1 taking a over b, v_2 taking b over c and v_3 taking c over a is already impossible.

We take Definition 1 as the definition, so part 1 of Theorem 2 is correspondingly stronger. For the 24-cell it verifies 786,432 *trees of which only* 62,208 *are reachable by any traversal: the other* 724,224 *are trees no implementation of breadth-first search will ever hand you.*

Remark 3 (The tree we compute)**.** *Selecting the parents requires a rule, and unless stated otherwise the tree we compute, written $\mathcal{T}_r$, is the one produced by the textbook BFS loop: cells are held in a FIFO queue, the queue is initialized with r, and each time a cell is dequeued its neighbors are scanned in increasing cell index, every neighbor not yet seen being marked, given the dequeued cell as its parent, and appended to the queue. The parent of v is thus whichever of its candidates entered the queue first.*

We spell this out because the rule is less innocent than it looks. The only arbitrary ingredient is the index order in which neighbors are scanned, but the queue order that decides parentage is derived from it recursively: cells at distance $d-1$ are dequeued in the order their own parents were, so the induced order is lexicographic along the discovery path rather than by index. It is therefore not the same as taking the smallest-indexed candidate. On the two polytopes of Section 4 for which the choice turns out to matter, the two rules disagree on 55% *and* 25% *of the cells that have more than one candidate. Note also that a traversal is not even able to reach every tree Definition 1 allows, by Remark 2; the trees enumerated for part 1 of Theorem 2 are chosen directly from the definition, not run through a traversal. Neither rule has any geometric content. For the polytopes of part 1 of Theorem 2 none of this matters, since* every *BFS tree is valid there; where it matters is Section 4.*

2.2 Face-Rotation BFS Algorithm

Given a 4-polytope P with vertex set $V \subset \mathbb{R}^4$, cell set C, and a root cell $r \in C$, the algorithm proceeds as follows.

Step 1: Face-up rotation. Let $\bar{V} = \{v - \bar{v} : v \in V\}$ where $\bar{v}$ is the centroid of V. Let $c_r = \text{centroid}(\{v \in \bar{V} : v \text{ is a vertex of } r\})$. Apply the rotation $R \in \text{SO}(4)$ that maps $c_r/\|c_r\|$ to the unit vector $\hat{w} = (0, 0, 0, 1)$. (If $c_r/\|c_r\| = \hat{w}$ we take $R = I$; if $c_r/\|c_r\| = -\hat{w}$ we negate all coordinates.) This *face-up* rotation ensures that the root cell's centroid lies on the positive w-axis, making subsequent unfolding into the hyperplane $\{w = \text{const}\}$ numerically stable.

Step 2: BFS traversal. Maintain a FIFO queue Q initialized with the root cell r, and a visited set $\text{Vis} = \{r\}$ with $T_r = \text{id}$. While Q is non-empty, dequeue the front cell p; for each neighbor c of p in $G(P)$, taken in the order they appear in the adjacency list, if $c \notin \text{Vis}$: add c to Vis, enqueue c, and compute the cumulative 4D affine transform T_c by the procedure in Algorithm 1. The algorithm terminates when Q is empty (no candidates remain); since $G(P)$ is connected for every convex

4-polytope, all cells are visited. The order in which neighbors are added to Q is the tie-breaking rule: it determines which of the BFS spanning trees of Definition 1 is produced (Lemma 1). In our implementation, ties are broken by the order in which neighbors appear in the adjacency list of the input data.

Step 3: Projection. Apply each T_c to the vertices of cell c, obtaining $\{\, T_c(v) \in \mathbb{R}^4 : v \in c \,\}$. After Step 2, the w-coordinate of every transformed vertex satisfies $|w - w_0| \le \varepsilon_w$ for a common value w_0 and numerical tolerance $\varepsilon_w \approx 10^{-13}$. Discard the w-coordinate; the resulting (x, y, z)-coordinates form the 3D net.

Algorithm 1 UnfoldTransform(p, c, T_p)

Require: parent cell p with cumulative transform T_p; child cell c
Ensure: cumulative transform T_c for c

Let $f_1, \dots, f_k$ be the vertices of the shared face of p and c (in the original 4D frame)
$\tilde{f}_i \leftarrow T_p(f_i)$ for all i ▷ shared face in parent's unfolded frame
$c_f \leftarrow \frac{1}{k} \sum_i \tilde{f}_i$ ▷ face centroid
Compute orthonormal basis $\{u_1, u_2\}$ of the face plane via Gram-Schmidt on $\{\tilde{f}_i - c_f\}$
$n_p \leftarrow$ component of $(T_p(\text{centroid}(p)) - c_f)$ orthogonal to $\{u_1, u_2\}$, normalized
$n_c \leftarrow$ component of $(T_p(\text{centroid}(c)) - c_f)$ orthogonal to $\{u_1, u_2\}$, normalized
$\theta \leftarrow \arccos\big(\max(-1, \min(1, n_c \cdot (-n_p)))\big)$
$e_1 \leftarrow n_c$; $\quad e_2 \leftarrow \text{normalize}(-n_p - (n_c \cdot (-n_p))\, n_c)$ ▷ if degenerate, use any vector $\perp \{u_1, u_2, n_c\}$
$R \leftarrow I_4 + (\cos\theta - 1)(e_1 e_1^\top + e_2 e_2^\top) + \sin\theta (e_2 e_1^\top - e_1 e_2^\top)$ ▷ 4D rotation in the plane $\{e_1, e_2\}$
return $\big(R,\ c_f - R\, c_f\big) \circ T_p$ ▷ compose affine maps

A naive implementation using *closed-region* membership (e.g. Mathematica's `RegionMember`) incorrectly classifies cells as overlapping when they share only a face, edge, or vertex in the net. This is particularly acute for pointed cells such as tetrahedra and octahedra, which frequently share edges and vertices with cells not connected by an edge of $\mathcal{T}_r$.

We use the following *volume-based* test, which detects only interior overlaps.

Remark 4. *Two cells not connected by an edge of $\mathcal{T}_r$ may share a face, edge, or vertex in the unfolded net without overlapping in their interiors. A correct validity test must detect only* interior *overlaps.*

Definition 2 (Valid net)**.** *A net is* valid *if for every pair of cells (i, j) not connected by an edge of $\mathcal{T}_r$,*

$$\mathrm{Vol}_3\big(\mathrm{conv}(c_i) \cap \mathrm{conv}(c_j)\big) \ \le\ \varepsilon_{\mathrm{vol}}$$

where $\mathrm{conv}(c_i)$ denotes the convex hull of cell i's vertices in 3D, and $\varepsilon_{\mathrm{vol}} = 10^{-8}$.

In practice we first apply a bounding-box filter to exclude pairs that clearly cannot overlap, then compute `RegionMeasure[RegionIntersection[`h_i`, `h_j`]]` in Mathematica for the remaining candidate pairs, where h_i denotes the convex hull mesh of cell i's vertices in 3D.

Remark 5. *Using closed-region membership instead of volume intersection would report the 16-cell, 24-cell, and 600-cell as invalid (false positives due to boundary contact), contradicting the correct result. The volume threshold $\varepsilon_{\mathrm{vol}} = 10^{-8}$ is well below the smallest possible cell volume (even for the 600-cell, where each tetrahedral cell has volume $\approx 3 \times 10^{-3}$ at the scale used), so no true interior overlap is missed.*

3 Results

3.1 Main Result

By Definition 1 the BFS trees rooted at r are in bijection with the parent choices, so there are exactly

$$\prod_{v \neq r} \left| N(v) \cap L_{d(v)-1} \right|$$

of them. This count mentions no rule, and lets us enumerate or sample the trees without reference to one. For the 5-cell the product is 1: there is only one BFS tree per root and a tie-breaking rule has nothing to decide. For the 8-, 16- and 24-cell it is 6, 20,736 and 32,768; for the 120- and 600-cell it is about 1.8×10^{44} and 1.8×10^{65}.

Theorem 2 (Main)**.** *Let P be a regular convex 4-polytope and r a cell of P.*

1. *If P is the 5-cell, 8-cell, 16-cell or 24-cell, then* every *BFS spanning tree rooted at r yields a valid net. In particular the conclusion does not depend on the tie-breaking rule.*

2. *If P is the 120-cell or the 600-cell, then the tree $\mathcal{T}_r$ of Remark 3 yields a valid net.*

Proof. Exhaustive computational verification. For each of the six polytopes we loaded the combinatorial data (vertices, faces, cells) from the data files available in the repository, unfolded from every root cell, and applied the volume-based intersection test (Definition 2) to every pair of cells not joined by an edge of the tree in question.

For part 1 we enumerated all BFS trees of every root: 5, 48, 331,776 and 786,432 trees for the 5-, 8-, 16- and 24-cell respectively. All are valid. For the 5-, 8- and 16-cell this also follows from [3], which covers all spanning trees; the 24-cell is new.

For part 2 the number of BFS trees puts enumeration out of reach, and we verified the tree $\mathcal{T}_r$ for each of the 120 and 600 roots. As evidence that the outcome is not an artifact of the rule, we additionally drew 20 BFS trees uniformly at random per root— 2,400 and 12,000 trees—and found all of them valid as well. Across all six polytopes 1,132,661 trees were checked without a single self-intersection.

Table 4 summarizes the results, and Figure 1 shows representative nets for each polytope. The w-coordinate spread after unfolding was at most 4×10^{-13} in all cases, confirming that all cells lie in a common 3D hyperplane to numerical precision. □

Remark 6 (How close the nets come to overlapping)**.** *The verification is numerical, and whether rounding could have hidden an overlap deserves a better answer than the value of ε_{vol}. We therefore measured, for every pair of cells not joined by an edge of $\mathcal{T}_r$ and every root, not whether the two cells meet but how far apart they are. For two convex polyhedra the direction realizing the minimum distance is a face normal of one of them or a cross product of an edge of each, so the largest gap over that finite set of directions* is *the distance; the computation is a few hundred inner products per pair. Cells are congruent, so only pairs whose centroids are close enough for the distance to be small need be examined.*

The verdict turns out to be nowhere near the boundary (Table 3). Every such pair either touches—sharing a face, an edge or a vertex, at distance zero to within 3×10^{-15}—or stands at least 0.086 *apart, and none lies in between. Touching is not an artifact of the arithmetic: the two cells carry common vertices of P, and the unfolding sends them to the same point. It is also common—from a quarter of the near pairs to all of them—and it is exactly what a closed-region membership test misreads. What matters is the width of the empty band between the two. About eleven orders*

of magnitude separate the error the unfolding accumulates, at most 4×10^{-13}*, from the smallest genuine gap, so no choice of* ε_{vol} *within ten orders of magnitude of ours changes a single verdict.*

Making the verification exact rather than merely robust would not mean redoing the unfolding. The floating-point run already exhibits, for each pair, a direction along which the two cells separate; certifying the result is a matter of re-evaluating those inner products in exact arithmetic, over $\mathbb{Q}$ *for the 8-, 16- and 24-cell and over* $\mathbb{Q}(\sqrt{5})$ *for the rest. The coordinates stay in these fields even though the hinge rotations involve* $\sin\theta$ *(for the 16-cell* $\theta = 60°$*): a regular cell is determined by one of its faces together with the side on which it lies, so each unfolded cell coincides, as a point set, with the reflection of its parent in the plane of their shared polygon, and a reflection can be written without normalizing its normal vector. The face-up rotation of Step 1 does introduce square roots, so an exact check would omit it and unfold into the root's own hyperplane. A separating hyperplane is the right certificate to ask for, since it proves what is actually wanted—that the interiors are disjoint—while permitting the contact that genuinely occurs.*

Table 3: How close the cells come to one another, over all roots, for every pair not joined by an edge of $\mathcal{T}_r$ whose centroids are near enough for the distance to be small. "Contact" is the largest deviation from zero among the pairs that touch, and "smallest gap" the least distance among those that do not. Distances are in the scale of the data files, for which the last column gives the diameter of a cell.

Polytope	Near pairs	Touching	Contact	Smallest gap	Cell diameter
5-cell	30	30	1.7×10^{-16}	—	1.73
8-cell	128	96	0	2.000	3.46
16-cell	1,536	1,248	3.3×10^{-16}	1.000	1.73
24-cell	2,760	1,368	8.9×10^{-16}	0.787	2.83
120-cell	38,940	10,047	1.6×10^{-15}	0.086	2.14
600-cell	5,780,237	2,064,332	2.7×10^{-15}	0.115	1.51

Table 4: Comparison of face-rotation BFS and apple-peel RZ unfolding. "Valid orderings" = fraction of (C_1, C_2) pairs for which apple-peel visits all cells; "Valid nets" = fraction of valid orderings that produce a non-self-intersecting 3D net.

		Face-rotation BFS	Apple-peel RZ [10]	
Polytope	Cells	Valid (all roots)	Valid orderings	Valid nets
5-cell	5	5/5	20/20 (100%)	20/20 (100%)
8-cell	8	8/8	48/48 (100%)	48/48 (100%)
16-cell	16	16/16	20/64 (31%)	20/20 (100%)
24-cell	24	24/24	192/192 (100%)	192/192 (100%)
120-cell	120	120/120	1440/1440 (100%)	**0/1440 (0%)**
600-cell	600	600/600	**0/2400 (0%)**	—

Remark 7. *For the 5-, 8-, and 16-cell, Theorem 2 is a special case of the stronger result of Devadoss and Harvey [3], who proved that* all *spanning trees of the cell-adjacency graph (not just BFS trees) yield valid nets. The new content of Theorem 2 is therefore the 24-cell, 120-cell, and 600-cell.*

Remark 8 (The BFS restriction is not vacuous)**.** *Call a polytope* all-net*, after O'Rourke, if* every *ridge unfolding of it is valid. Devadoss and Harvey prove that the 600-cell is not all-net ([3], Theorem 11) and conjecture the same for the 24-cell and the 120-cell. Both conjectures have since been confirmed: the 24-cell by Seamons [6], and all three together by Devadoss, Harvey and Richter [4], who exhibit an overlapping chain of facets in each. The classification is therefore complete—of the six regular convex 4-polytopes the 5-, 8- and 16-cell are all-net and the 24-, 120- and 600-cell are not—and it lines up exactly with ours: the three on which Theorem 2 is new are precisely the three that fail.*

What separates the two verdicts is the restriction to BFS trees, and the restriction is not a technicality: none of the known counterexamples is a BFS tree, nor can any be completed to one.

For the 24-cell this is immediate. The chain of [4] runs through 14 octahedra, a path of 13 edges, whereas the cell-adjacency graph has eccentricity 3 *from every cell, so a BFS tree has depth* 3 *and its longest path has at most* 6 *edges. Seamons's spanning tree fails for a weaker version of the same reason:* 13 *of its* 24 *cells sit deeper than their distance from the root.*

For the 120-cell we checked it. A path lies in a BFS tree rooted at r exactly when the distances from r along it change by one at each step and have no interior maximum: a path in a rooted tree climbs to an apex and descends, and a cell with both neighbors nearer r would need two parents. Their chain of nine dodecahedra fails this test from every one of the 120 *roots.*

For the 600-cell we checked more, because their chain of eight tetrahedra shares a vertex and so lies in a single vertex figure, whose 20 *cells carry the graph of a dodecahedron. That graph has* 1,440 *paths on eight cells. Unfolding every one of them,* 120 *overlap and* 600 *lie in some BFS tree, and no path does both. Whichever of them [4] exhibit, it is not part of a BFS tree.*

Remark 9 (The 8-cell has only one net)**.** *For the 8-cell the theorem admits a shorter proof, and the net is a familiar one. Two cells of the 8-cell are adjacent unless they are opposite, so the layers from any root are $1+6+1$: the root, its six neighbors, and the opposite cell, whose parent may be any of the six. The unfolding places the root at the origin, its six neighbors on the six faces of that cube, and the last cube beyond one of them. That is Dalí's cross, and the six trees give it in the six possible orientations, so all* 48 *nets of the 8-cell are congruent.*

Remark 10 (On the number of verifications)**.** *Each regular convex* 4*-polytope has a symmetry group acting transitively on cells. A symmetry g carrying a root r to a root r' induces an automorphism of $G(P)$ carrying $L_k(r)$ to $L_k(r')$, hence a bijection between the BFS trees rooted at r and those rooted at r'; and the unfolding is equivariant, the net obtained from $(r', g(T))$ being congruent to the one obtained from (r, T), so that one is valid exactly when the other is.*

For part 1 a single root would therefore have sufficed, and no assumption about the tie-breaking rule is needed: the statement quantifies over all *BFS trees of the root, and that whole family is carried onto the corresponding family at any other root. We ran every root regardless, the extra cost being negligible, and no root behaved differently from any other.*

Part 2 is not of this form, since it names one tree per root rather than all of them. The tree $\mathcal{T}_r$ of Remark 3 is selected by cell index, which is not equivariant, so $\mathcal{T}_{r'}$ need not be $g(\mathcal{T}_r)$ and validity at one root implies nothing about another; the 120 *and* 600 *roots were accordingly verified individually. Whether* every *BFS tree is valid for these two polytopes—which part 1 settles for the other four—remains open.*

3.2 Contrast with Apple-Peel Unfolding

The apple-peel algorithm [10] constructs a spanning tree by greedily selecting the next cell according to a global spiral (RS) or zonal (RZ) criterion, without the layer-by-layer structure of BFS. Table 4

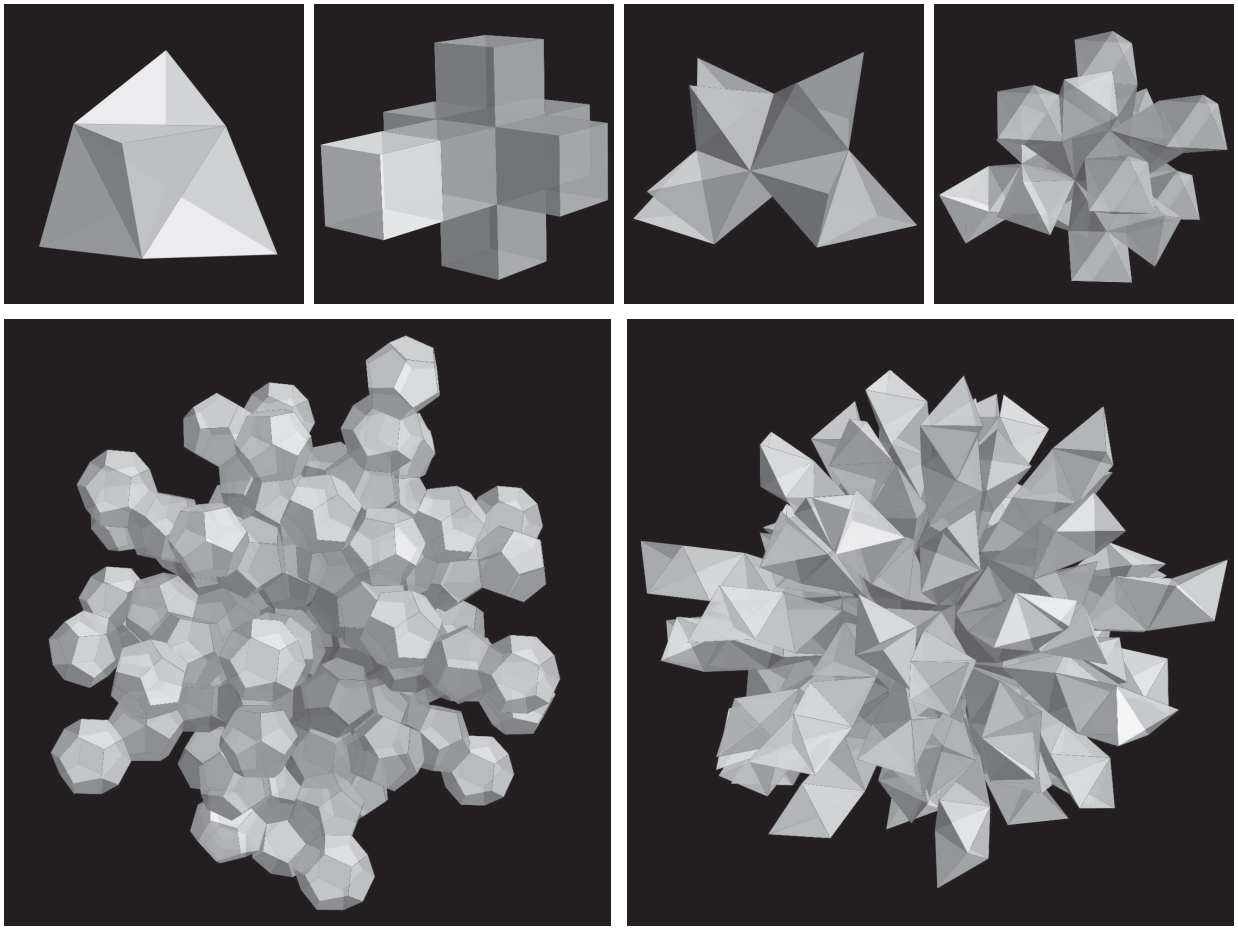


Figure 1: Face-rotation BFS nets of the six regular convex 4-polytopes. Top row (left to right): 5-cell, 8-cell, 16-cell, 24-cell. Bottom row: 120-cell, 600-cell. Shading indicates BFS layer depth (dark = root cell, light = periphery). STL files of all six nets are available in the repository (see Code and Data Availability).

compares the two algorithms.
The most striking contrast occurs for the 120-cell and 600-cell:

- **120-cell**: Apple-peel produces 1,440 valid orderings (all (C_1, C_2) pairs succeed), yet every resulting 3D net self-intersects [10]. Face-rotation BFS yields a valid net from every root.
- **600-cell**: Apple-peel cannot complete any ordering (structural impossibility due to the 4-regular cell-adjacency graph). Face-rotation BFS yields a valid net from every root.

These cases demonstrate that *the existence of a valid ordering and the validity of the resulting 3D net are independent properties.* A spanning tree that is well-suited for the ordering problem need not yield a geometrically valid net, and vice versa.

4 The Uniform Case

Theorem 2 does not extend to the convex uniform 4-polytopes. Excluding the two infinite prismatic families, there are exactly 64 of these; unlike the enumeration of all uniform 4-polytopes, which is still open, this list is known to be complete [2, 5], the exceptional non-Wythoffian member being the grand antiprism of Conway and Guy [2]. We generated all 64 and ran the face-rotation BFS unfolding from every cell, 24,487 roots in all.

Construction. The 45 distinct Wythoffian members of the A_4, B_4, F_4 and H_4 families were built from their Coxeter diagrams: the group is generated as the closure of the four mirror reflections, the vertices are the orbit of the point whose distance to each ringed mirror is 1 and to each unringed mirror 0 (the edge generated by mirror i has length twice that distance, so equal distances force equal edge lengths), and the d-faces are the orbits of the rank-d parabolic subgroups. The snub 24-cell and the grand antiprism are not Wythoffian in this sense, but both arise as diminishings of the 600-cell: delete an inscribed 24-cell, respectively two orthogonal decagonal rings, and take the convex hull. In each case the surviving cells are the 600-cell tetrahedra avoiding every deleted vertex, together with one cell per deleted vertex, namely the hull of its remaining neighbors. The 17 prisms are the prisms over the 5 Platonic and 13 Archimedean solids, with lateral edge length equal to the edge length of the base, minus the cube prism, which is the tesseract and already regular. Each construction was checked against the group-theoretic predictions for the vertex and cell counts, the Euler relation $V - E + F - C = 0$, the requirement that every ridge lie in exactly two cells, and vertex-transitivity of the vertex degree.

Results. The outcome splits three ways, as summarized in Table 5.
Throughout this paragraph the tree is $\mathcal{T}_r$; how much of what follows survives a change of rule is taken up below. Sixty-one of the 64 yield a valid net from every root. The runcinated 24-cell $t_{0,3}\{3,4,3\}$ yields one from no root: all 240 of its unfoldings self-intersect, in between 3 and 23 pairs of cells. This last is a statement about $\mathcal{T}_r$ and not about the polytope; a tree chosen equivariantly does give it a valid net from every root, as we show at the end of this section. The remaining two are *root-dependent*: the runcitruncated 600-cell $t_{0,2,3}\{5,3,3\}$ succeeds from 47 of its 2,640 roots and the runcitruncated 120-cell $t_{0,1,3}\{5,3,3\}$ from 2,570 of 2,640. Figure 2 shows one failure of each kind.

The two root-dependent cases are the informative ones. For every regular 4-polytope the choice of root is immaterial, and it is tempting to read that as a consequence of cell-transitivity. The uniform case shows that once cell-transitivity is dropped, validity genuinely depends on where the

Table 5: Face-rotation BFS nets of the 64 convex uniform 4-polytopes, over all root cells, for the tree $\mathcal{T}_r$ of Remark 3. The columns count roots, not trees, and the classification is one of (polytope, rule) pairs rather than of polytopes: under other rules four of the 61 in the first column lose roots and one of them, the grand antiprism, loses all of them. See Table 6.

Family	Polytopes	Roots	All valid	Root-dependent	None valid
A_4, B_4, F_4	30	1,893	29	0	1
H_4	15	21,600	13	2	0
Snub 24-cell, grand antiprism	2	464	2	0	0
Prisms	17	530	17	0	0
Total	64	24,487	61	2	1

unfolding starts: admitting *some* valid face-rotation BFS net and admitting one from *every* root are different properties, and Theorem 2 establishes the stronger one only in the regular case.

Cell degree does not separate the three outcomes, in either direction. The grand antiprism and the snub 24-cell both contain cells of degree 4—the same poverty of connection that makes the 600-cell impossible for apple-peel unfolding—yet both are valid from every root, while the runcinated 24-cell has minimum cell degree 5 and is valid from none. Nor does high degree help: every prism has two cells adjacent to all of the others, up to degree 92 for the prism over the snub dodecahedron, and all 17 are valid from every root, whereas the root-dependent runcitruncated 120-cell also carries cells of degree 32.

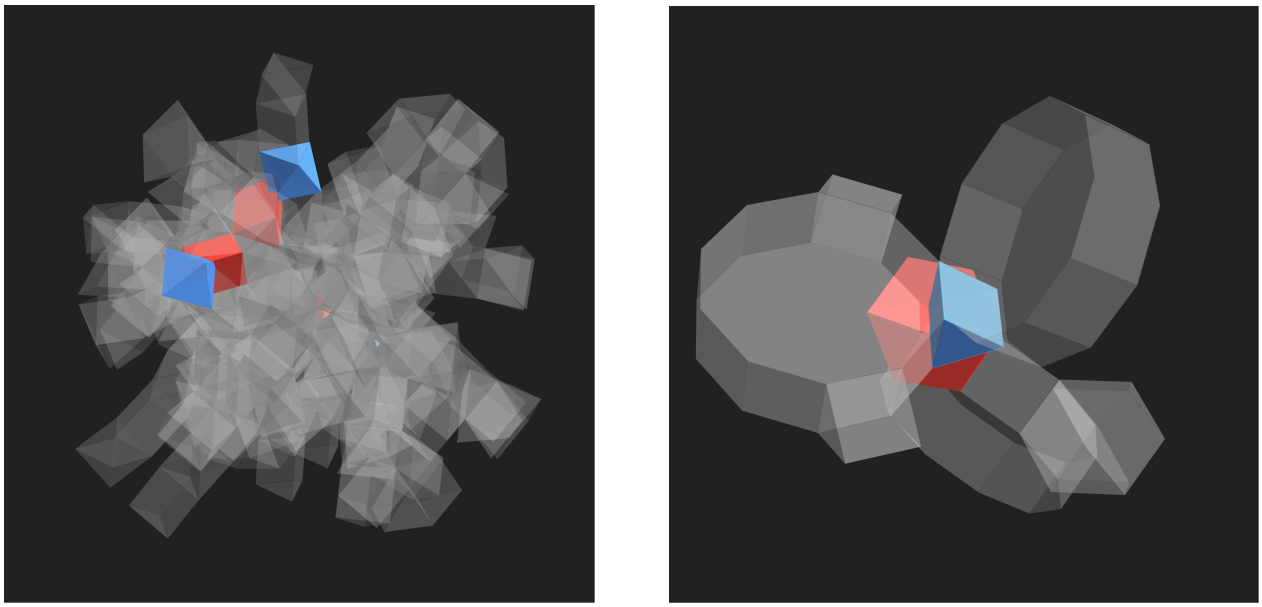
The classification is a statement about one rule. Table 5 counts the tree $\mathcal{T}_r$ of Remark 3, one per root. That is not the same as counting BFS trees, and for the uniform polytopes the difference is not a technicality: the classification moves when the rule does. Table 6 gives the three exceptional polytopes under three rules—the queue of Remark 3, and choosing for each cell the candidate parent of largest or smallest index.

Table 6: Roots that yield a valid net, over all roots, under three rules for choosing parents.

Polytope		queue	max-index	min-index
Runcinated 24-cell	$t_{0,3}\{3,4,3\}$	0/240	0/240	0/240
Runcitruncated 600-cell	$t_{0,2,3}\{5,3,3\}$	47/2640	2635/2640	2627/2640
Runcitruncated 120-cell	$t_{0,1,3}\{5,3,3\}$	2570/2640	2640/2640	2151/2640
Grand antiprism		320/320	320/320	0/320

The runcitruncated 600-cell goes from 1.8% of its roots to 99.8%; the runcitruncated 120-cell reaches every root under max-index. The grand antiprism is the sharpest case, and it is not one of the three: it is valid from all 320 roots under the queue and under max-index, and from *none* under min-index. So a polytope can sit in any of the three classes depending on how parents are chosen, and "valid from every root" is a property of a rule together with a polytope, not of the polytope. Three more of the 61 lose roots under one of these rules, though only a few each: $t_{0,2,3}\{3,3,3\}$ loses one of its 30 under max-index, and $t_{0,3}\{4,3,3\}$ and $t_{0,3}\{5,3,3\}$ lose one of 80 and 26 of 2,640 under min-index.

Sampling BFS trees uniformly instead of fixing a rule separates the uniform polytopes from the regular ones more sharply still. Ten random trees at one root leave 48 of the 61 untouched, but 13 fail, and the failures are concentrated among the large polytopes: every one of the four polytopes



(a) (b)

Figure 2: The two ways the unfolding fails. The panels are at deliberately different scales, because the claims they carry are about different things: (a) is about a polytope, (b) about a single root. (a) The runcinated 24-cell at its *best* root, drawn whole: 3 of its 240 cells intersect a partner, the fewest over all 240 roots, and no root does better than 3. (b) The runcitruncated 120-cell at one of its 70 failing roots, drawn as a close-up of the single intersecting pair together with its neighbors; the other 2,570 roots are fine. In both panels the two members of an intersecting pair are drawn in different colors.

with 2,640 cells fails all ten—two of them being the root-dependent pair, and so outside the 61 to begin with—as does the grand antiprism, which is valid from all 320 of its roots under the queue rule and under max-index. Against this, the regular polytopes did not produce a single failure in 1,132,661 trees. What distinguishes the regular case is not that some tree works but that the choice does not matter, and it is exactly that insensitivity which the uniform polytopes lack.

A net where no rule of Table 6 finds one. Every rule considered so far is index-based, and none of them is equivariant under the symmetry group. Asking instead for a tree that *is* equivariant changes the verdict on the runcinated 24-cell. Its cell-adjacency graph has automorphism group of order 2,304 — the extended group $[3,4,3]$, the diagram $x3o4o3x$ being palindromic — acting with two orbits of cells, of sizes 48 and 192. Fix a root r and let H be a subgroup of its stabilizer. Because the unfolding is equivariant, a BFS tree invariant under H produces a net invariant under H; such a tree exists exactly when every cell has a candidate parent fixed by its stabilizer in H, and for a root in either orbit the largest such H has order 3. Sampling 600 of these invariant trees at a root of each orbit gave 70 and 83 valid nets respectively. The symmetry group is transitive on each orbit and carries valid nets to valid nets, so the runcinated 24-cell has a valid face-rotation BFS net from *every* one of its 240 root cells.

The control is what makes this meaningful: 600 unconstrained random BFS trees at each of the same two roots produced no valid net at all. The gain comes from the symmetry, not merely from abandoning the index rules. The runcinated 24-cell is therefore not a polytope without a net, but a polytope whose nets no index-based rule finds — the sharpest form of the conclusion of the

previous paragraph.

Caveats. For the polytopes with more than a few hundred cells the intersection test was carried out with a separating-axis implementation rather than by measuring the volume of the intersection; the two were checked to return identical verdicts, and identical numbers of intersecting pairs, on every one of the 1,893 roots of the 30 polytopes of the A_4, B_4 and F_4 families, and again on a sample of roots of the two root-dependent cases, where the verdict is also stable as the separation tolerance ranges over 10^{-4} to 10^{-10}.

5 Open Problems

The main open question raised by this work is:

> *Why does face-rotation BFS always produce a valid net for regular 4-polytopes? Is there a theoretical proof that does not rely on computer enumeration?*

A second question is whether Theorem 2 extends to all spanning trees and not merely to BFS trees, as [3] proved for the 5-, 8-, and 16-cell. It does not, and for none of the three: by Remark 8 each of them admits an overlapping spanning tree. What is left is the question in between. Part 1 answers it for the 24-cell, where every BFS tree is valid; for the 120- and the 600-cell we have verified $\mathcal{T}_r$ from every root and a sample beyond it, and whether *every* BFS tree of those two is valid is open. With 1.8×10^{44} trees per root—and 1.8×10^{65} for the 600-cell—enumeration is hopeless and a proof would have to be structural rather than computational.

A third question is the characterization left open by Section 4. Under a fixed rule the 64 uniform 4-polytopes split three ways, and what we can offer is a list of quantities that do *not* decide the split. Cell degree does not, in either direction. Neither does the angular defect around an edge—2π less the sum of the dihedral angles that the cells meeting the edge contribute there—although it is a tempting candidate. A defect of zero would mean that those cells tile $\mathbb{R}^3$ face-to-face around the edge, so the defect measures how far the polytope stands from a Euclidean honeycomb, and convexity forces it to be positive. For a regular polytope it is a single number, and it orders the six exactly by how hard they have proved to be: 148.4°, 90°, 77.9°, 31.6°, 10.3° and 7.4° for the 5-, 8-, 16-, 24-, 120- and 600-cell. Among the uniform polytopes it fails in both directions. The omnitruncated 120-cell has the same minimum defect, 0.736°, as the root-dependent runcitruncated 120-cell, and is valid from every root; the runcinated 24-cell sits at 10.5°, above many polytopes that are valid from every root. Weighting the defect by the depth or by the size of the polytope does not repair this, and it costs the one clean signal the raw defect had: the six regular polytopes no longer come out in order. Whether the split is captured by any local quantity at all, rather than by some global property of how the unfolding folds back on itself, we do not know.

A related question is how far the counts depend on the tie-breaking rule. Section 4 shows that they depend on it a great deal, and that the dependence is not monotone in any obvious way: the runcinated 24-cell is valid from no root under three index rules and from every root under an equivariant one. Whether root-dependence survives every choice of rule for the two remaining cases is open.

Finally, extensions to higher dimensions are open as well.

Code and Data Availability

All Mathematica scripts implementing the face-rotation BFS algorithm, the volume-based intersection test, and the visualization are available at `https://github.com/takashi-randomwalker/apple-peel-4d`. The same repository contains the constructions used for Section 4—the Wythoff generator, the two diminishings of the 600-cell, and the prisms—and the separating-axis implementation of the intersection test used for the larger polytopes, together with its agreement check against the volume-based test. It also contains the combinatorial data of all 64 convex uniform 4-polytopes, the nets exported as STL and OFF, and the verdict for each of the 24,487 individual roots. STL files of the face-rotation BFS nets for all six polytopes (root cell 1) are also provided in the same repository, as are the two scripts that decide whether the overlapping chains of [4] can lie in a BFS tree.

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